

Instagram Redesign

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Abstract— Since its creation in October 2010, Instagram has become one of the most used social media platforms in the world. Over the years, Instagram has continually added more and more functionality for its users in addition to its original design of only having users post pictures. One of the most fast growing features that Instagram offers is the ability to post customized videos known as reels. The main task of this paper is to offer an alternative interface design for Instagram’s current user interface for watching reels.

1 INSTAGRAM REEL

Instagram reels has been a popular format for video posts on the application for a number of years. Reels were first developed and added into Instagram in August 2020 (Instagram, 2020). Since its deployment, Instagram reels have become a prevalent part of user consumption of social media posts in the app.

1.1 Description

Instagram has many features very interconnected and mainly revolves around the use of posting social media content. The content can range from pictures to videos. Instagram reels is a derivative of videos that are posted on the app that usually follows the convention of being 60 seconds or less and has a variety of possible effects, sounds, edits, or filters. Depending on the type of post, Instagram has a different display interface for each type. In addition, the display design for some types can also be the same. For example, the display for reels and photos can be the same like the figure below. The interface for the viewing reels shown in the middle image is the main way users watch reels while using Instagram. The design of this interface mainly consists of the video itself. The reel encompasses a majority of the screen with specific buttons overlaid on top of the video for the user’s convenience. The only section of the screen the reel left is the bottom section of the screen for the Instagram main task bar. The overlaid functions on top of the reel for the user includes a button for the task of returning back to the previous page, like, share, comment, and menu of other options. The

interface also utilizes many popular techniques for touch screen applications such as double tap to like and scroll up and down to move between posts.



Figure 1—Instagram interface display: pictures or video (left), reels (middle), profile reel page (right). Source: [Image](#)

1.2 Access

Instagram is a completely free application for everyone to be able to use. The only caveat to this is that the user must be at least 13 years old or the required age depending on the region. To access Instagram and begin viewing reels, you must create an Instagram account. First you will need to download Instagram on your mobile device. Once the download is finished, opening the app will prompt you to either login or create a new account. Creating the account will require an email address. Next, you must enter the confirmation code that will be sent to your email. This will give you access to Instagram where you will then need to enter your birthday and username for the account. With all of these steps done, the only thing left is to click on the magnifying glass at the bottom main task bar. This will redirect you to a search page in the app with a large wall of random posts that can be scrolled through. This page will have a large variety of reels to browse through, and clicking on any of these reels will allow you to view them.

2 HEURISTIC EVALUATION

Like any interface design, there are factors that lead an interface to be good or bad. Instagram's reel watching interface has many good features that improves its user ability to complete his or her task of watching reels. However, it also has many negative aspects to the design as well.

2.1 What Works Well

Instagram's current reel watching design has a very basic design that focuses solely on allowing the target to watch the reel. The mobile application gives the majority of the area of the screen towards the video itself so that the user is able to watch the video with the best quality. While giving the user a majority of the screen real estate towards the video, the interface includes a dedicated area for the major tasks of Instagram as well. The user only loses a small section of the screen at the bottom while having constant access to the application's most important menu. In addition, the decision to overlay the button of different tasks in the display also helps give the user the ability to see what other options for the video are possible without losing space on the screen. The developers of this design also decided to follow along with typical mobile traditions such as scrolling up and down to access other posts to aid the user in the consistency of mobile use. With this in mind, the user also has multiple options to do the same task like liking videos by double tapping the screen or clicking on the heart button.

2.2 Why It Is Good

All of the aforementioned factors contribute to the positive effects that the current design has. By using the surface area of the screen effectively, the user is able to have a majority of his or her focus on the video which is the main goal of the interface. Not eliminating the main task bar completely or hiding the taskbar would have caused a major issue with the navigation of the overall application so by minimizing the task bar and maximizing the reel video area, the user is able to enjoy the reels to the fullest without losing functionality to the rest of the application. The overlaying of the buttons also contributes to this factor as well. Since the buttons are layered directly on top of the video, video related functionality is kept intact by simply sacrificing a small area of the video. Even so, the reels are clearly visible through the opaque design of the buttons. Most mobile apps follow standard touch screen designs which allows for easy usage

throughout all the apps that use the standard. Instagrams follows this standard as well, which ultimately helps the user have an easier time using the application without having distinctive buttons telling the user how to do basic tasks, like scrolling up and down to move through posts. The interface also gives users buttons for more obscure functionality that may not be as intuitive to users such as double tapping the screen to like a post.

2.3 Good Principles

Instagram's interface design follows many of the heuristic design principles that allows the interface to be well received. Among these include, structure, simplicity, mapping, consistency, and flexibility. The structure of design is the most important aspect that mainly gives the design its positive features. The design of the interface itself not only gives clear and concise purpose to every portion of the display, but also makes each section easily recognizable for the users of the app. Since the reel is the main focus of the task, the design itself purposely allocates a large portion of the screen towards this goal. It also gives adequate spacing for buttons of all the possible minor tasks that the user may encounter such as navigation or commenting on a post. This design clearly adheres well to the principle of structure. Simplicity also plays a role in Instagram's reels. The display keeps all the design extremely basic for the user and focuses solely on the video itself. It only allocates buttons towards the most probable actions a user may do while using the interface, while any extra options the user might do can be found in a given menu button. The mapping of the buttons are also very explicit and helps the user in understanding the interface as well. Each button adheres to a specific design that almost always clearly explains what the button does without having to explicitly tell the user. The mapping of the button designs works well with the simplicity of the interface to help users complete his or her task. Consistency and flexibility are two heuristic principles that are interconnected with the current design. While Instagram's reel watching display both has the characteristics of consistency and flexibility, they are evenly balanced to give the user the benefits of both. The design allows the user to follow the typical touch screen standards of mobile apps like scrolling up and down to see other posts and also double tapping to like a post. However, it also allows more inexperienced users to use the heart button to like a post as well. Both consistency and flexibility can be seen in the reel design. All of these principles contribute to the positive elements of the original design due to the increase in invisibility of the interface. The combination of design principles in

the current interface gives a seamless transition to all the different tasks that a user may come across while using the interface while still giving main focus towards the overall task of watching the reel. Direct manipulation is another aspect of the interface that attributes to the increased invisibility of the application. By being able to scroll in multiple directions to be able to control the app, the user is able to focus solely on the task itself. Instagram's design effectively helps greatly narrow the gulf of execution for the user by specifically targeting the identify actions and execute phases through the use of these good design principles.

2.4 What Does Not Work Well

Although Instagram's current reel interface works very well and hardly has any negative aspects, there are still some elements that cause conflict for the user to complete his or her tasks. One of the issues that arises for users is the inability to control reels well. The nature of reels are short videos that users can watch and enjoy. However, when viewing a reel, the reel automatically starts playing once a user clicks on it. This may not seem like an issue; however, the part of the interface that is affected by this is the current scroll bar for the video. If a user desires to skip ahead or backtrack in the video, this task can be daunting. The current design has the scroll bar extremely small and barely visible. In addition, the only pause functionally available for reels is to press and hold on the screen. There is no indefinite pause button available. In addition to this issue, Instagram's reel navigation has very poor human error tolerance. Hypothetically, if a user is watching a video and would like to rewatch the video, the user must scroll up to see the old video again. However, if the user accidentally scrolls to the right instead of up, this will automatically take the user back to the search page and the old video will be lost.

2.5 Why It is Bad

Instagram's main usage is for the avid consumption of mass social media posts. However, with the current interface design users are unable to properly control the reels that are watched or avoid any human errors that cause posts to be missed. The video controls of the reels have two major flaws that contribute to its negative effect. The first flaw surrounds the size and usability of the video scroll bar. Since the bar is barely visible and extremely small with no other indications that it is there, unless a user is already familiar with the available functionality, a

user will never be able to take advantage of this option. In addition, with its small touch screen area allocation, the usability of this functionality drops significantly since users will have a difficult time maneuvering the video bar. This leads into the other issue with the current design which is the lack of human error tolerance. Reels that have been completely finished playing will automatically be pushed up and the next reel will immediately play. This gives the user little opportunity to be able to complete any other task on the finished reel. If a user tries to go back to the previous reel, the only way is to swipe down on the application to go back. Since this action is very similar to the swipe right functionality, any user can accidentally trigger the wrong action and navigate back to the search page. If this occurs, the user will lose access to the reel that he or she desired to look at again, ultimately resulting in the task failing.

2.6 Bad Principles

The design problems of the reel video scroll bar and human error stems from the lack of the heuristics of discoverability, ease, and tolerance in the current interface. The lack of these elements directly affects the usability of the interface. Since the video scroll bar's design gives this option for users a very low chance of discovering this functionality, the chances of the bar being used at all drops significantly. The bar's design purposely makes the bar extremely small to see as well as making the color of the bar opaque. With this in mind, this severely hinders a user's ability to identify the bar as a usable functionality in the interface to aid in tasks. In addition, the design of this bar also affects the ease and comfort of the interface. The size of the bar also comes with the difficulty in using the bar itself. Even though a person's finger may be small and can easily tap accurately on a screen, with such a small area to touch to activate the bar, any user can struggle attempting to use the reel scrolling even if the user is aware it is there. Human error tolerance is a vital part of any interface to assist users; however, Instagram does not give the user much leeway in its use, especially towards navigation and control. Since posts play automatically and control in the reel watching display utilizes scrolling, any slight mistakes the user makes can result in the user skipping or losing posts that he or she may otherwise want to watch or revisit. Although the current design helps tremendously with narrowing the gulf of execution in many aspects, there are still flaws in the design that widens the gulf of execution instead. All of these bad design principles hinders the user's

identify actions and execution phase of the gulf of execution. This will in turn cause the user to have difficulty in completing his or her tasks.

3 INTERFACE REDESIGN

3.1 Prototype Design Features/Changes

The prototype design will focus on alleviating the two issues that can be found in the current interface. This will be done with changes to two aspects of the current interface design and one addition to the design

1. Instagram reel video scrolling will be made larger and constantly visible with associated video buttons
2. The scroll/flick to the right function to navigate to the previous page will include an confirmation for accidental scroll/flicks
3. Instagram reel videos will not automatically move on and play the next reel once the current reel is finished playing

3.2 Paper Prototype Visual Mock Up

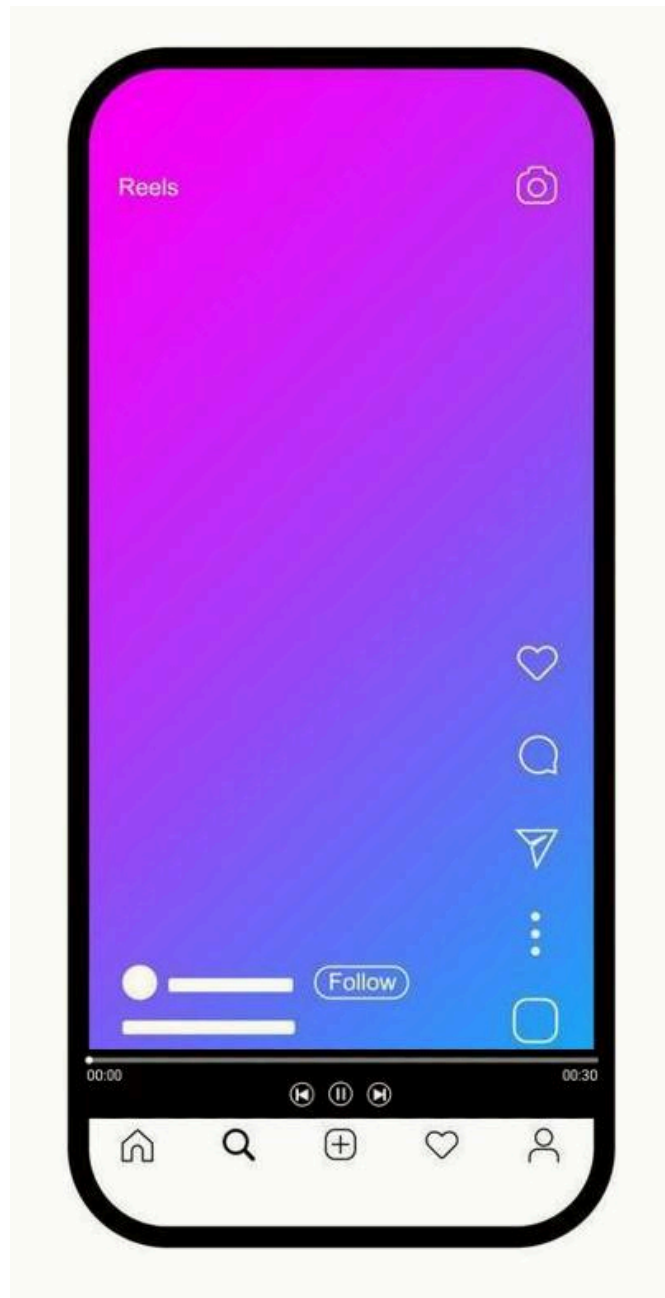


Figure 2—Prototype with scroll bar and associated buttons.
Original unaltered image source: [Image](#)

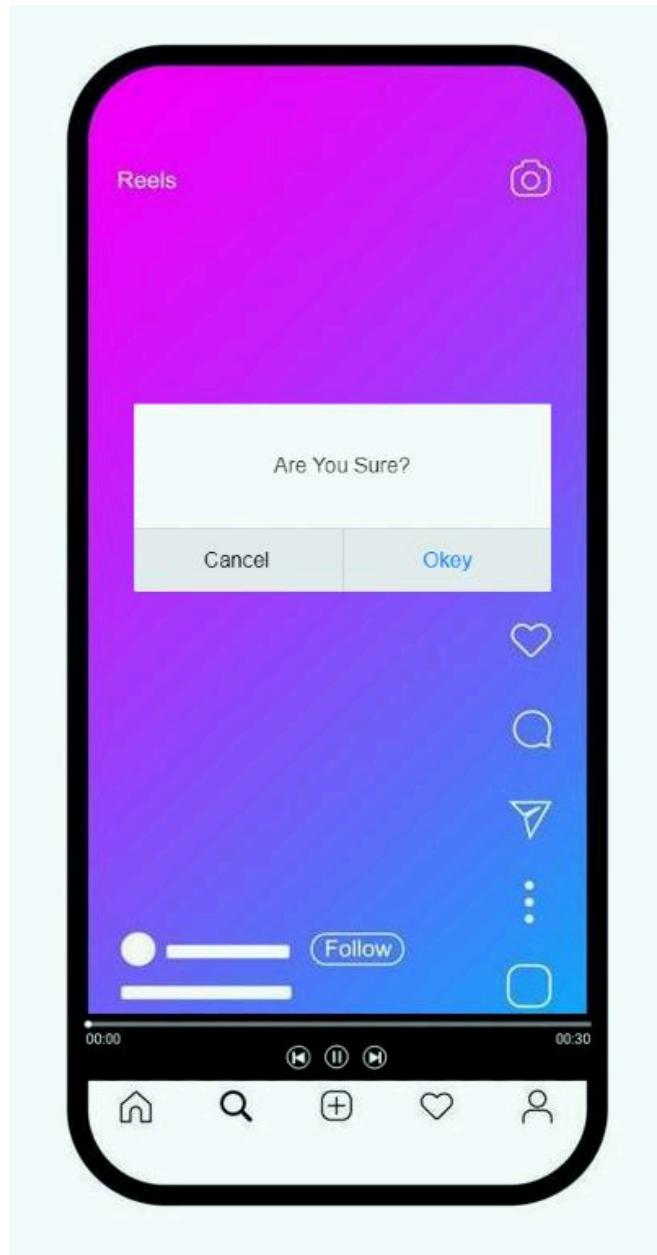


Figure 3—Prototype with right scroll/flick navigation confirmation.
Original unaltered image source: [Image](#)

4 INTERFACE JUSTIFICATION

The designs shown are specifically implemented towards improving the bad principles that the current interface has. The design principles that are improved is the discoverability, ease, and tolerance of the reef watching display.

4.1 Redesign Justification

The three features of the redesign each has an individual purpose that targets one of the three design principles being improved. The first change to the original design recreates the reel video scrolling bar by making the bar a part of the bottom of the reef. Although the image does not signify this, the bar is meant to follow the same design as the buttons to be overlaid onto of the video so that the viewing of the reef is not hindered. The bar will be made much bigger and more prominent than before to clearly signify its role to the users. In addition, the buttons added will also follow the same design. The buttons will also give the user complete control over the reel and give the users more usability for options involving reel itself. The second change from the original design slightly alters the flow of the reels being played. Currently the next reel will automatically play after the current reel is finished; however, the new design will simply stop the reel giving control back to the user. The user will be given the opportunity to do any tasks or action on the current reel that he or she desires before moving onto the next. To be able to move onto the reel, the scrolling functionality will be kept the same. The last change the new design incorporates is the use of confirmation in navigation when the user utilizes the right scrolling. This confirmation will only be displayed through this horizontal scrolling and not when the user presses the back button on the reel which has the same functionality. This gives the user the option for more precise actions and avoids accidental navigational scrolling.

4.2 Improvements to Bad Principles

Since each feature specifically targets each design principle with flaws in the original interface, improvements to the bad design principles are clear. With the more pronounced and clear scroll bar, the discoverability of this feature is greatly heightened. With this change, users will be able to quickly identify the availability and use of this feature in the application and be able to better identify how to complete his or her task involving the scroll bar. In addition, with the larger size of the scroll bar, the user's ease of use will be improved as well. Since

the bar's size directly affects the touch screen use capabilities, users will be able to much easily manipulate the bar to his or her liking. This will result in the usability of the interface to improve along with improvements of the reel video scrolling. The included buttons also directly affect the ease and comfort of the user. By giving more control of the interface back to the user, the interface's ease of use will greatly be increased. Another change that also gives the user more control of the application and assists with ease and comfort is the change in the design where reels are played automatically. Since the application now has less control over the reels, the user is able to decide how the next task is done, whether it is to comment or to play the next reel. Human error tolerance is the last principle that is greatly affected by the redesign. Since the use of scrolling is prevalent with mobile apps, changing any of its functionality will cause consistency issues. To avoid this, adding in a confirmation for the right scrolling navigation in the reels watching interface allows the user to completely eliminate any errors coming from accidental scrolling. This gives the user an extra step in going back to the search page which will force the user to lose access to any unsaved posts he or she just saw. With this extra step, the user is able to determine whether or not any errors have occurred and respond accordingly. Although this does not completely eliminate all human error in the interface, this will greatly alleviate one of the biggest human errors coming from Instagram's reel interface.

4.3 Preservation of Good Principles

With the addition of the new interface designs and changes, it is impossible to ensure that all the good design principles that the current interface has is preserved. However, since the new design focuses solely on the bad design principles, most of the heuristics of the original are kept intact. The only design principles that were slightly affected were structure and simplicity. Although the structure of the design was slightly changed, it was a minor change that simply added in a large scroll bar. This ultimately should not have a major negative effect on the overall design. Simplicity of the interface should be slightly lowered with the addition of the confirmation popup for the navigational scrolling. Nevertheless, this change should also only have a minimal effect on the good design principles of the current interface. Overall, the structure, simplicity, mapping, consistency, and flexibility of the application should all still have a positive effect on the interface. The structure and simplicity of the designs were only minorly changed to better assist with major flaws within the design

principle so this will balance out. The mapping of the buttons of the interface were left completely unchanged and will not have any negative repercussions within the new design. The redesign follows along with consistency and purposely utilizes the error popup to allow consistency within mobile applications to continue instead of altering the functionality of scrolling itself and causing inconsistency for the app. The flexibility of the reel interface is kept completely intact due to the application already having multiple processes for the same action. The horizontal scroll has the exact same result as the button to go back which means that the popup does not affect flexibility since the user still has access to a non-popup way of navigating. All of the original design's strengths have been properly preserved to the best possible with the new design.

4.4 Overall

With the improvements to the lack of key heuristics from the new design and the preservation of the good qualities of the original design, Instagram's reel interface design is now able to maximize the user's ability to complete his or her task. With the addition of more discoverability, ease, and human error tolerance, the gulf of execution will become narrower and the improvements of the interface are highly probable.

5 REFERENCES

1. Instagram. (2020, August 5). *Introducing Instagram Reels*. Instagram.
<https://about.instagram.com/blog/announcements/introducing-instagram-reels-announcement>